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United States Patent	12398031
Kind Code	B2
Date of Patent	August 26, 2025
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Micro-device structures with etch holes

Abstract

A micro-device structure comprises a source substrate having a sacrificial layer comprising a sacrificial portion adjacent to an anchor portion, a micro-device disposed completely over the sacrificial portion, the micro-device having a top side opposite the sacrificial portion and a bottom side adjacent to the sacrificial portion and comprising an etch hole that extends through the micro-device from the top side to the bottom side, and a tether that physically connects the micro-device to the anchor portion. A micro-device structure comprises a micro-device disposed on a target substrate. Micro-devices can be any one or more of an antenna, a micro-heater, a power device, a MEMs device, and a micro-fluidic reservoir.

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Family ID: **1000008777747**

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Appl. No.: **18/646280**

Filed: **April 25, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250042717 A1	Feb. 06, 2025

Related U.S. Application Data

continuation parent-doc US 17475046 20210914 US 12006205 child-doc US 18646280
continuation-in-part parent-doc US 17066448 20201008 US 11952266 20240409 child-doc US 17475046
us-provisional-application US 63173988 20210412

Publication Classification

Int. Cl.: **B81B1/00** (20060101); **H10N30/30** (20230101)

U.S. Cl.:

CPC **B81B1/004** (20130101); **H10N30/306** (20230201); B81B2201/058 (20130101); B81B2203/0118 (20130101); B81B2203/0307 (20130101); B81B2203/0353 (20130101)

Field of Classification Search

CPC: B81B (1/004); B81B (2201/058); B81B (2203/0118); B81B (2203/0307); B81B (2203/0353); H10N (30/306); B81C (1/00476)

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Background/Summary

PRIORITY APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/475,046, filed on Sep. 14, 2021, now issued as U.S. Pat. No. 12,006,205, which claims the benefit of U.S. patent application Ser. No. 17/066,448, filed on Oct. 8, 2020, now issued as U.S. Pat. No. 11,952,266, and U.S. Provisional Patent Application No. 63/173,988, filed on Apr. 12, 2021, the disclosure of each of which is hereby incorporated by reference herein in its entirety.

CROSS REFERENCE TO RELATED APPLICATIONS
(1) The present disclosure is related to U.S. patent application Ser. No. 17/006,498, entitled Non-Linear Tethers for Suspended Devices by Trindade et al., filed Aug. 26, 2020, and to U.S. patent

application Ser. No. 17/066,448, entitled Micro-Device Structures with Etch Holes by Rubino, filed Oct. 8, 2020, the disclosure of each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The present disclosure generally relates to micro-transfer printed devices and structures that enable device release from a source wafer.

BACKGROUND

(3) Components can be transferred from a source wafer to a target substrate using micro-transfer printing. Methods for transferring small, active components from one substrate to another are described in U.S. Pat. Nos. 7,943,491, 8,039,847, and 7,622,367. In these approaches, small integrated circuits are formed on a native semiconductor source wafer. The small, unpackaged integrated circuits, or chiplets, are released from the native source wafer by pattern-wise etching sacrificial portions of a sacrificial layer located beneath the chiplets, leaving each chiplet suspended over an etched sacrificial portion by a tether physically connecting the chiplet to an anchor separating the etched sacrificial layer portions. A viscoelastic stamp is pressed against the process side of the chiplets on the native source wafer, adhering each chiplet to an individual stamp post. The stamp with the adhered chiplets is removed from the native source wafer. The chiplets on the stamp posts are then pressed against a non-native target substrate or backplane with the stamp and adhered to the target substrate. In another example, U.S. Pat. No. 8,722,458 entitled Optical Systems Fabricated by Printing-Based Assembly teaches transferring light-emitting, light-sensing, or light-collecting semiconductor elements from a wafer substrate to a target substrate or backplane.

(4) Crystalline source wafers can etch anisotropically so that the etch proceeds more rapidly in one direction than in another direction with respect to the crystal planes in the source wafer. Thus, a sacrificial portion of the source wafer can etch more rapidly in one direction than another so that a micro-device or tether disposed over the sacrificial portion is only partially undercut and are partially unetched. Despite the difference in etch rates in different directions in a crystalline source wafer, micro-devices, tethers, and anchor portions can still be attacked by the etchant, especially if the etch takes a substantial amount of time, compromising the integrity of the micro-device, tether, and the anchor portion and possibly leading to failures in micro-device pickup from the source wafer with the stamp and failures in micro-device performance on the target substrate. Furthermore, slow etch rates can reduce manufacturing throughput and, because the etch chemistry selectivity amongst materials is not infinite, the encapsulating layers should be exposed as little as possible to the wet etch chemistry as they also etch away (albeit at much slower rates).

(5) There is a need, therefore, for micro-device and source wafer structures and methods that facilitate micro-device release from a source wafer in less time and with reduced etching damage to micro-devices and transfer structures.

SUMMARY

(6) The present disclosure provides, inter alia, structures and methods for improving the release of micro-devices from a source substrate to enable micro-transfer printing the micro-devices from the source substrate to a target substrate. According to some embodiments, a micro-device structure comprises a source substrate having a sacrificial layer comprising a sacrificial portion adjacent to one or more anchor portions (e.g., one or more anchors) and a micro-device disposed completely over the sacrificial portion. The sacrificial portion can be the substrate itself or a gap formed by etching a sacrificial portion of material, for example. The micro-device can have a top side opposite the sacrificial portion and a bottom side adjacent to the sacrificial portion and the micro-device can comprise an etch hole that extends through the micro-device from the top side to the bottom side. A tether can physically connect the micro-device to the anchor portion. According to some embodiments, the etch hole is in contact with the sacrificial portion and exposes at least a portion of the sacrificial portion, the etch hole is not at a geometric center of the micro-device, the etch hole is at the geometric center of the micro-device, the etch hole is rectangular, the etch hole

has a portion of a perimeter that is at the geometric center, or any compatible combination of these.

(7) According to embodiments of the present disclosure, the sacrificial portion of the source substrate has a crystalline structure that is anisotropically etchable, the source substrate is made of silicon {100}, the source substrate has a crystalline structure with a {100} orientation (such that a {100} crystal plane is exposed at a surface of the source substrate) and the bottom side of micro-device is substantially parallel to the source substrate surface and to a {100} crystal plane on the surface, or any combination of these. According to some embodiments, the micro-device has a micro-device edge direction oriented at an angle from 0 to 90 degrees, and in some embodiments 30 to 60 degrees, with respect to the {110} crystal plane. The micro-device can have a micro-device edge direction oriented at an angle of substantially (e.g., within manufacturing tolerances) 45 degrees with respect to the {110} crystal plane. The source substrate can have a crystalline structure with a {100} crystal plane, the micro-device can have a micro-device width and the direction of the micro-device width can be substantially (e.g., within manufacturing tolerances) parallel to the {100} crystal plane and oriented from 0 to 90 degrees and more specifically 30 to 60 degrees, such as 45 degrees, with respect to a {110} crystal plane. The etch-hole width can be measured in a direction parallel to the tether width and can be longer than the etch-hole length.

(8) According to some embodiments, the etch hole is rectangular and has an etch-hole width, the tether has a tether width connecting the micro-device to the anchor portion, and the etch-hole width is no less than the tether width. That is, the extent of the connection between the micro-device and the tether (or the anchor portion and the tether) in a direction is equal to or smaller than the extent of the etch hole in the same direction, e.g., in a direction parallel to the connection between the micro-device and the tether or to the connection between the anchor portion and the tether. In some embodiments, the etch-hole width is greater than the tether width. The etch hole and the micro-device can both be rectangular. The etch hole can have an etch-hole edge, the micro-device can have a micro-device edge, and the etch-hole edge can be substantially parallel to the micro-device edge. The etch hole can have an etch-hole edge with an etch-hole width in the range of 5 microns to 20 microns. The micro-device can have a micro-device length no greater than 5 mm, 2 mm, 1 mm, 500 microns, 200 microns, 150 microns, or 100 microns.

(9) According to some embodiments, the tether physically connects the anchor portion directly to a corner of the micro-device. According to some embodiments, a micro-device structure comprises micro-devices, each micro-device disposed completely over a corresponding sacrificial portion in a direction orthogonal to the source substrate surface and physically connected by a respective tether to the anchor portion, wherein each of the micro-devices and the respective tether are rotated with respect to any other of the micro-devices and the respective tether. The anchor portion can have a surface that is a square or circle. The anchor portion can substantially be a cube or cylinder, or an equivalent structure.

(10) According to embodiments of the present disclosure, a micro-device structure comprises a target substrate, a micro-device disposed on or over the target substrate, the micro-device having a top side and a bottom side opposite the top side and adjacent to the target substrate, and comprising one or more etch holes that extend through the micro-device from the top side to the bottom side, and at least a portion of a broken (e.g., fractured) or separated tether physically connected to the micro-device. The etch hole can have an etch-hole edge with an etch-hole width in the range of 5 microns to 20 microns. The micro-device can have a micro-device length no greater than 5 mm, 2 mm, 1 mm, 500 microns, 200 microns, 150 microns, or 100 microns.

(11) According to some embodiments of the present disclosure, a micro-device structure comprises a source substrate having a sacrificial layer comprising one or more sacrificial portions each adjacent to an anchor portion, micro-devices, each disposed completely over one of the one or more sacrificial portions, and a respective tether for each of the micro-devices. Each of the micro-devices is physically connected to the anchor portion by the respective tether. In some embodiments, each of the micro-devices and the respective physically connected tether are rotated

or reflected, or both reflected and rotated, with respect to each other of the micro-devices and the respective physically connected tether. In some embodiments, the respective tether for each of the micro-devices physically connects the anchor portion directly to a different corner of the micro-device or to a portion of the micro-device closer to the corner than to a center or opposite corner along an edge of the micro-device.

(12) According to some embodiments of the present disclosure, the micro-device is, comprises, or provides one or more of an antenna, a micro-heater, a power device, a MEMs device, and a micro-fluidic reservoir.

(13) According to some embodiments of the present disclosure, a method of making a micro-device structure comprises providing a source substrate comprising a sacrificial layer (e.g., a portion or layer of the source substrate) comprising or providing a sacrificial portion adjacent to an anchor portion, providing a micro-device disposed completely over the sacrificial portion, the micro-device having a top side opposite the sacrificial layer and a bottom side adjacent to the sacrificial layer and comprising an etch hole that extends through the micro-device from the top side to the bottom side and a tether that physically connects the micro-device to the anchor portion, wherein the etch hole exposes a portion of the sacrificial portion, providing an etchant, and etching the sacrificial portion, wherein at least the exposed portion of the sacrificial portion is etched by the etchant passing through the etch hole, thereby forming a gap between the micro-device and the source substrate such that the micro-device is suspended from the anchor portion by the tether, for example over the source substrate or over a target substrate.

(14) Methods of the present disclosure can comprise providing a stamp and a target substrate, contacting the micro-device with the stamp to adhere the micro-device to the stamp, removing the stamp from the source substrate, thereby breaking (e.g., fracturing) or separating the tether, pressing the micro-device to a target substrate to adhere the micro-device to the target substrate, and removing the stamp. The etchant can be TMAH or KOH.

(15) According to some embodiments of the present disclosure, an intermediate structure wafer comprises a source substrate having a sacrificial layer comprising a sacrificial portion adjacent to an anchor portion and an intermediate substrate disposed completely over the sacrificial portion. The intermediate substrate has a top side opposite the sacrificial portion and a bottom side adjacent to the sacrificial portion and comprises an etch hole that extends through the intermediate substrate from the top side to the bottom side. One or more micro-devices are disposed on the intermediate substrate (e.g., the top side of the intermediate substrate). The one or more micro-devices are non-native to the intermediate substrate. A tether physically connects the intermediate substrate to the anchor portion.

(16) In some embodiments, one of the one or more micro-devices comprises a micro-device comprising a micro-device etch hole aligned with the etch hole in the intermediate substrate.

(17) In some embodiments, the one or more micro-devices is a plurality of micro-devices that are electrically, optically, or both electrically and optically interconnected.

(18) According to embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side, one or more etch holes, an anchor, and one or more tethers. Each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side. In some embodiments, at least one etch hole has an aspect ratio no less than 2:1. The one or more tethers physically connect the micro-device to the anchor.

(19) According to embodiments of the present disclosure, any straight line parallel to an edge side of the micro-device drawn from a concave corner on a first edge side of the micro-device to a second edge side of the micro-device opposite the first edge side contacts (e.g., intersects or touches) an etch hole. Concave corners can be defined by the micro-device alone or by the intersection of a tether with the micro-device. An edge side is a side of the micro-device that is on a perimeter of the micro-device and does not correspond to a top side or bottom side of the micro-

device. An edge side can be a micro-device edge and can be a line segment portion of a micro-device having a polygonal perimeter. Opposing or opposite sides (edge sides) are edge sides that do not contact and are not adjacent to each other, for example opposite edge sides of a rectangular micro-device.

(20) According to embodiments of the present disclosure, any orthogonal line segment angle drawn from a first concave corner on a first edge side of the micro-device to a second concave corner on a second edge side of the micro-device adjacent to the first edge side contacts (e.g., intersects or touches) an etch hole.

(21) According to embodiments of the present disclosure, at least one etch hole has two or more portions that extend in different directions and is disposed closer to a center of the micro-device than to an edge side of the micro-device.

(22) According to embodiments of the present disclosure, the micro-device is unitary and contiguous. The anchor can surround the micro-device, for example the anchor can surround all of the edge sides of the micro-device, can extend around a perimeter of the micro-device, or surrounds two or more edge sides around the perimeter of the micro-device. In some embodiments, no etch hole intersects an edge side of the micro-device. In some embodiments, one or more etch holes intersect an edge of the micro-device. At least one etch hole can be disposed at an angle that is not parallel or orthogonal to an edge side of the micro-device or at least one etch hole can form an x (‘x’) shape, a plus (‘+’) shape, a T (‘T’) shape, a Y (‘Y’) shape, or a cross shape.

(23) According to embodiments of the present disclosure, a micro-device structure comprises a source substrate, the anchor is disposed on or is a portion of the source substrate, and the micro-device is separated from the source wafer by a cavity. The source substrate can have a crystalline structure that is anisotropically etchable. The source substrate can be made of silicon, for example silicon {100} or silicon {111}. According to some embodiments, the source substrate has a crystalline structure with a {100} orientation and the bottom side of the micro-device is substantially parallel to a {100} crystal plane at a surface of the source substrate. According to some embodiments, the micro-device has a micro-device edge direction (e.g., edge side direction) oriented at an angle from 30 to 60 degrees with respect to a {110} crystal plane of the crystalline structure. According to some embodiments, the micro-device comprises a micro-device edge (e.g., edge side) having a direction oriented at an angle of substantially 45 degrees with respect to the {110} crystal plane. According to some embodiments, the source substrate has a crystalline structure with a {100} crystal plane, the micro-device has a micro-device length greater than a micro-device width, and the micro-device length is in a direction that is substantially parallel to the {100} crystal plane.

(24) According to some embodiments of the present disclosure, at least one etch hole and the micro-device (e.g., excluding any tethers) are both rectangular. In some embodiments the etch hole has an etch-hole edge, the micro-device has a micro-device edge, and the etch-hole edge is substantially parallel to, substantially orthogonal to, or substantially at 45 degrees to the micro-device edge.

(25) According to some embodiments of the present disclosure, one or more tethers (e.g., a plurality of tethers) physically connect one edge side (e.g., only one edge side) of the micro-device to the anchor. In some micro-device structures of the present disclosure, the micro-device can have two or more edge sides and can comprise one or more tethers (e.g., multiple tethers) physically connecting two or more edge sides of the micro-device to the anchor or to multiple anchors. The micro-device structure can comprise piezoelectric material and the micro-device can be or comprise a mass that mechanically stresses the piezoelectric material when mechanically perturbed.

(26) According to some embodiments of the present disclosure, a micro-device system comprises a target substrate, one or more micro-device structures disposed on the target substrate, and a fractured or separated structure tether physically attached to the anchor.

(27) According to some embodiments of the present disclosure, a micro-device system comprises a

source substrate, one or more micro-device structures native to and disposed on the source substrate, and a structure tether (e.g., a fractured or separated tether) physically attached to the anchor.

(28) According to some embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side and comprising two or more etch holes, wherein each of the two or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, an anchor, and one or more tethers physically connecting the micro-device to the anchor.

(29) According to some embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side, one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, an anchor, and two or more tethers physically connecting the micro-device to the anchor.

(30) According to some embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side, two or more concave corners, one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, an anchor, and wherein any straight line parallel to an edge side of the micro-device drawn from a concave corner on a first edge side of the micro-device to a second edge side of the micro-device opposite the first edge side contacts an etch hole.

(31) According to some embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side, one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, an anchor, and one or more tethers physically connecting the micro-device to the anchor, and wherein any orthogonal line segment angle drawn from a first concave corner on a first edge side of the micro-device to a second concave corner on a second edge side of the micro-device adjacent to the first edge side contacts an etch hole.

(32) According to some embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side, one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, an anchor, and one or more tethers physically connecting the micro-device to the anchor, and wherein at least one etch hole has two or more portions that extend in different directions and is disposed closer to a center of the micro-device than to an edge side of the micro-device.

(33) According to embodiments of the present disclosure, a micro-device structure comprises a substrate comprising a sacrificial portion disposed in or on the substrate, a micro-device disposed entirely on the sacrificial portion, wherein (i) the micro-device has at least one of a length and a width no less than 100 μm (e.g., no less than 200 μm, 300 μm, 400 μm, 500 μm, 750 μm, or 1 mm) and a top side and a bottom side and (ii) comprises one or more etch holes and each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, one or more anchors disposed on the substrate, and one or more tethers physically connecting the micro-device to the one or more anchors, wherein the one or more etch holes are sized, shaped, and oriented relative to the sacrificial portion such that the micro-device can be completely released by etching the sacrificial portion thereby suspending the micro-device over the substrate by only the one or more tethers (e.g., within two hours, one hour, 30 minutes, or 15 minutes of etching).

(34) According to embodiments of the present disclosure, a micro-device structure comprises a substrate comprising a sacrificial portion disposed in or on the substrate, a micro-device disposed entirely on the sacrificial portion, wherein (i) the micro-device has at least one of a length and a width no less than 100 μm (e.g., no less than 200 μm, 300 μm, 400 μm, 500 μm, 750 μm, or 1 mm) and a top side and a bottom side and (ii) comprises one or more etch holes and each of the one or

more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, one or more anchors disposed on the substrate, and one or more tethers physically connecting the micro-device to the one or more anchors, wherein the one or more etch holes are sized, shaped, and oriented relative to the sacrificial portion such that no pinned etch front forms when etching the sacrificial portion.

(35) According to embodiments of the present disclosure, a micro-device having a top side and a bottom side comprises one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, wherein the one or more etch holes comprise a first etch hole extending into the micro-device from a first edge side, a second etch hole extending into the micro-device from a second edge side that is opposite the first edge side, and a third etch hole extending into the micro-device from a third edge side different from the first edge side and from the second edge side, wherein the third etch hole comprises a first portion that intersects the third edge side and a second portion that intersects an end of the first portion at a point other than an end of the second portion. The first portion can bisect the second portion, the first etch hole and the second etch hole can be parallel, the first etch hole and the second etch hole can be collinear, or the first portion can extend into the micro-device past the first etch hole and the second etch hole.

(36) According to embodiments of the present disclosure, a micro-device having a top side and a bottom side comprises one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, and wherein the one or more etch holes comprise a plurality of etch holes disposed in a symmetric arrangement, the symmetric arrangement having a multifold symmetry and comprising a unit cell comprising a first portion of a first etch hole extending outward from a center of the micro-device, a second etch hole that is an interior etch hole and disposed parallel to an edge side of the micro-device, and a third etch hole that is an interior etch hole and that is co-linear with a center of the micro-device. The symmetric arrangement can have two-fold or four fold symmetry.

(37) According to embodiments of the present disclosure, the first etch hole has a cross shape. The second etch hole and the third etch hole can be straight etch holes. The third etch hole can be collinear with both the center of the micro-device and a corner of the micro-device.

(38) According to embodiments of the present disclosure, a micro-device having a top side and a bottom side comprises one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side, and wherein the one or more etch holes comprise a first etch hole comprising a first portion and a second portion, wherein the first portion and the second portion intersect at respective interior points and a center of the first etch hole is disposed closer to a center of the micro-device than any edge side of the micro-device. The first portion and the second portion can each intersect the center of the micro-device. The first portion and the second portion can form a cross. The first portion can have a length equal to a length of the second portion. The first etch hole can be the only etch hole.

(39) According to embodiments of the present disclosure, a micro-device structure comprises a micro-device, one or more anchors, and one or more tethers physically connecting the micro-device to the one or more anchors. Embodiments of the present disclosure can comprise a source substrate comprising a sacrificial portion disposed in or on the source substrate, wherein the one or more anchors are disposed on the source substrate and the micro-device is native to and disposed entirely on the sacrificial portion.

(40) According to embodiments of the present disclosure, a micro-device structure comprises a micro-device having a top side and a bottom side and one or more etch holes. Each of the one or more etch holes is an etch hole that extends through the micro-device from the top side to the bottom side. An etch-hole protective coating comprising a material different from a material comprising the micro-device is disposed on the one or more etch holes (e.g., on sides of the one or more etch holes). The micro-device structure further comprises one or more anchors and one or

more tethers physically connecting the micro-device to the one or more anchors. The etch-hole protective coating can also be disposed on a top side, on a bottom side, or on both the top side and the bottom side of the micro-device. The micro-device can be disposed over a sacrificial portion and the micro-device and the sacrificial portion can comprise a same material, e.g., silicon. In some embodiments, the micro-device comprises KNN.
(41) Structures and methods described herein enable an efficient, effective, and fast release of a micro-transfer printable device or component from a source substrate (e.g., a native source wafer on or in which the device is disposed or formed) with reduced etching damage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing and other objects, aspects, features, and advantages of the present disclosure will become more apparent and better understood by referring to the following description taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1**A is a schematic plan view of a source wafer according to illustrative embodiments of the present disclosure;
(3) FIG. **1**B is a schematic detail cross section of a micro-device on the source wafer taken across cross-section line A of FIG. **1**A according to illustrative embodiments of the present disclosure;
(4) FIG. **1**C is a schematic detail cross section of a micro-device on the source wafer taken across cross-section line B of FIG. **1**A according to illustrative embodiments of the present disclosure;
(5) FIG. **1**D is a schematic detail cut-away perspective of a micro-device on the source wafer corresponding to FIGS. **1**A-**1**C according to illustrative embodiments of the present disclosure;
(6) FIG. **1**E is a detail perspective of a micro-device on the source wafer corresponding to FIGS. **1**A-**1**C according to illustrative embodiments of the present disclosure;
(7) FIG. **2**A is a schematic cross section of a micro-device on the source wafer after etching taken across cross-section line A of FIG. **1**A according to illustrative embodiments of the present disclosure;
(8) FIG. **2**B is a schematic cut-away perspective of a micro-device on the source wafer corresponding to FIG. **2**A according to illustrative embodiments of the present disclosure; and
(9) FIG. **3**A is a schematic plan view of micro-devices on a source substrate according to illustrative embodiments of the present disclosure;
(10) FIG. **3**B is a perspective of a micro-device on a source substrate corresponding to FIG. **3**A according to illustrative embodiments of the present disclosure;
(11) FIG. **4** is a cross section of micro-devices on a target substrate corresponding according to illustrative embodiments of the present disclosure;
(12) FIG. **5** is a flow diagram according to illustrative embodiments of the present disclosure;
(13) FIG. **6** is a graph illustrating release simulation results for various tether widths and etch-hole widths according to illustrative embodiments of the present disclosure;
(14) FIG. **7** is a perspective of an antenna micro-device according to illustrative embodiments of the present disclosure;
(15) FIG. **8** is a perspective of a micro-heater micro-device according to illustrative embodiments of the present disclosure;
(16) FIG. **9** is a perspective of a micro-fluidic reservoir structure micro-device according to illustrative embodiments of the present disclosure;
(17) FIG. **10** is a perspective of a piezo-electric power micro-device according to illustrative embodiments of the present disclosure;
(18) FIGS. **11**A-B are a schematic cross section and perspective view, respectively, of an intermediate structure wafer according to illustrative embodiments of the present disclosure;

(19) FIG. **12** is a perspective of a micro-device structure having two or more tethers and two or more etch holes suspended over a source wafer according to illustrative embodiments of the present disclosure;
(20) FIG. **13** is a perspective of a micro-device structure having two or more tethers on each edge side of a rectangular micro-device and two or more etch holes suspended over a source wafer according to illustrative embodiments of the present disclosure;
(21) FIG. **14** is a plan view of a micro-device structure having one tether on each edge side of a rectangular micro-device and four etch holes with extended arms according to illustrative embodiments of the present disclosure;
(22) FIG. **15**A is a plan view of the micro-device structure of FIG. **14** indicating some pinned etch fronts and concave corners, FIG. **15**B is a plan view showing straight line segments, and FIG. **15**C is a plan view showing orthogonal line segment angles according to illustrative embodiments of the present disclosure;
(23) FIGS. **16-18** are plan views of a micro-device with multiple tethers on a single edge side of the micro-device and multiple high-aspect-ratio etch holes according to illustrative embodiments of the present disclosure;
(24) FIGS. **19-20** are plan views of a micro-device with multiple tethers on multiple edge sides of the micro-device and multiple high-aspect-ratio etch holes according to illustrative embodiments of the present disclosure;
(25) FIG. **21**A is a plan view of a micro-device, tethers, and etch holes, FIG. **21**B is a plan view of the micro-device, tethers, and etch holes of FIG. **21**A indicating some pinned etch fronts, and FIG. **21**C is a perspective of the micro-device, tethers, and etch holes of FIG. **21**A according to illustrative embodiments of the present disclosure;
(26) FIGS. **22-23** are plan views of micro-device structures having a single etch hole with extended arms according to illustrative embodiments of the present disclosure;
(27) FIG. **24** is a cross section of a micro-device system according to illustrative embodiments of the present disclosure; and
(28) FIGS. **25**A and **25**B are cross sections and FIG. **25**C is a plan view with a detail of an etch hole with an etch hole protective coating in a micro-device system according to illustrative embodiments of the present disclosure.
(29) Features and advantages of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. The figures are not drawn to scale since the variation in size of various elements in the Figures is too great to permit depiction to scale.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(30) The present disclosure provides, inter alia, micro-device and source substrate (wafer) structures and methods that facilitate micro-device release from a source substrate in less time and with reduced etching damage to micro-devices and to tethering and anchor structures of the micro-device and source substrate, thereby facilitating the efficient construction, release, and micro-transfer printing of functional micro-devices from a source substrate. A source substrate can be a source wafer.
(31) According to some embodiments of the present disclosure and as shown in the plan view of FIG. **1**A, the detail cross section of FIG. **1**B taken across cross section line A of FIG. **1**A, the detail cross section of FIG. **1**C taken across cross section line B of FIG. **1**A, and the detail perspectives of FIGS. **1**D and **1**E corresponding to FIG. **1**A, a micro-device structure **90** comprises a source substrate **10** (e.g., a source wafer) having a sacrificial layer **12** comprising a sacrificial portion **14** adjacent to an anchor portion **16**. A micro-device **20** is disposed completely over sacrificial portion **14**. Micro-device **20** has a top side **24** opposite sacrificial layer **12** and sacrificial portion **14** and a

bottom side **26** adjacent to sacrificial layer **12** and sacrificial portion **14**. An etch hole **30** extends through micro-device **20** from top side **24** to bottom side **26** and exposes and contacts sacrificial portion **14**, providing access to sacrificial portion **14** through etch hole **30** by an etchant. Etch hole **30** has an etch-hole width W.sub.E. A tether **40** having a tether width W.sub.T physically connects micro-device **20** to anchor portion **16**. Tether width W.sub.T and etch-hole width W.sub.E can be parallel to each other and can be in a direction parallel to an edge of anchor portion **16** adjacent to micro-device **20** or an edge of micro-device **20** adjacent to anchor portion **16** where tether **40** connects micro-device **20** to anchor portion **16** and parallel to source substrate surface **11**, as shown in FIGS. **1**C and **1**D. As illustrated for example in FIG. **1**A, edge-hole width W.sub.E can be greater than a length of edge hole **30** in a direction orthogonal to edge-hole width W.sub.E and edge-hole width W.sub.E can be in a direction parallel to a length L.sub.M of micro-device **20**. Source substrate **10** has a source substrate surface **11** and a flat edge (source substrate flat **18**) used to orient and position source substrate **10** for photolithographic processing, as is typical in the photolithographic arts. Source substrate flat **18** is usually specified and oriented with respect to the crystallographic structure of source substrate **10**. In some embodiments, a location, orientation, or size of source substrate flat **18** indicates a doping characteristic of source substrate **10**, for example p-type or n-type doping. Micro-device **20** has one or more micro-device edges **22** (e.g., micro-device edge sides **22**). In some embodiments, micro-device **20** is rectangular and a micro-device edge **22** extends in a direction D parallel to source substrate surface **11** and at substantially 45 degrees to source substrate flat **18** (e.g., within manufacturing tolerances). In perspective cut-away FIG. **1**D, source substrate **10** surrounds sacrificial portion **14** in a common plane, as indicated with dashed lines. FIG. **1**E illustrates source substrate **10** extending to the common plane on which is disposed micro-device **20**. As used herein, micro-device edge **22** is an edge side **22**.

(32) Etch hole **30** can be, but is not necessarily, centered at or aligns with (e.g., an edge of etch hole **30** aligns with) a geometric center of micro-device **20**. Etch hole **30** can be, but is not necessarily, rectangular (e.g., square), and can have an etch-hole width W.sub.E(e.g., any extent) parallel to source substrate surface **11**. Micro-device **20** can be protected by patterned dielectric structure(s) **50**. Dielectric structure(s) **50** can be a part of micro-device **20** (e.g., an encapsulation layer), or a separate structure.

(33) According to some embodiments of the present disclosure, source substrate **10** has a crystalline structure that is anisotropically etchable. For example, source substrate **10** can be a semiconductor or compound semiconductor substrate or have a semiconductor or compound semiconductor sacrificial layer **12** disposed on a substrate. In some embodiments, source substrate **10** is a silicon substrate, e.g., comprising monocrystalline silicon, such as silicon {100} or silicon {111}. Micro-device **20** can be constructed in an epitaxial layer of source substrate **10** using conventional photolithographic methods and materials. According to some embodiments and as shown in FIG. **1**A, source substrate **10** has a crystalline structure with a {100} orientation (parallel to source substrate surface **11**) and bottom side **26** (a bottom surface) of micro-device **20** is substantially parallel (e.g., within manufacturing tolerance) to source substrate surface **11** and the {100} crystal plane of source substrate **10**. Likewise, ignoring any topographical structure formed by photolithographic processing of micro-device **20**, top side **24** (a top surface) of micro-device **20** can be substantially parallel (e.g., within manufacturing tolerances) to bottom side **26**, source substrate surface **11**, and the {100} crystal plane of source substrate **10**. Source substrate **10** has a source substrate flat **18** (e.g., a wafer flat that provides a specified orientation and alignment for source substrate **10**) that aligns with the {110} crystal plane of source substrate **10**. According to some embodiments, micro-device edge **22** direction D is oriented at an angle from 0 to 90 degrees and more specifically 30 to 60 degrees with respect to the {110} crystal plane, for example substantially at 45 degrees. Thus, assuming that source substrate surface **11** is in an x and y plane and that a z axis is perpendicular to source substrate surface **11** and the x and y planes, micro-device **20** can generally be rotated about the z axis with respect to source substrate flat **18** and

{110} crystal planes and, in some embodiments has a particular rotation relative to the z axis to have an edge at substantially 45 degrees to a {110}crystal plane. According to some embodiments, micro-device **20** has a micro-device length L.sub.M greater than a micro-device width W.sub.M and the direction of micro-device length L.sub.M is substantially parallel to the {100} crystal plane and can be oriented at an angle from 0 to 90 degrees with respect to the {110} crystal plane, for example substantially 45 degrees, e.g., in direction D. In some embodiments, micro-device **20** is rectangular. In some embodiments, micro-device **20** is square or has a non-rectangular shape.

(34) According to some embodiments of the present disclosure and as illustrated in FIGS. **1**A-**1**D, etch hole **30** can be rectangular, can have an etch-hole edge **32**, can have an etch-hole width W.sub.E, or any combination of these. Etch hole **30** can have tapered side walls so that a top side (top opening) of etch hole **30** has a greater area than a bottom side (bottom opening) of etch hole **30** adjacent to source substrate **10**. Tether **40** can have a tether width W.sub.T extending along an edge side **22** of micro-device **20** and connecting micro-device **20** to anchor portion **16**. Etch-hole width W.sub.E can be greater than or no less than tether width W.sub.T. By providing etch-hole width W.sub.E greater than or no less than tether width W.sub.T, sacrificial portion **14** is more readily etched with an etchant beneath micro-device **20** so that micro-device **20** and tether **40** are undercut before the etchant significantly damages anchor portion **16**, micro-device **20**, or both, as shown in the cross section of FIG. **2**A and perspective of FIG. **2**B. According to some embodiments of the present disclosure and as illustrated in FIGS. **2**A-**2**B, sacrificial portion **14** becomes a gap **15** (e.g., a cavity) formed by etching (e.g., wet etching or dry etching) or otherwise removing the material comprising sacrificial portion **14**, so that micro-device **20** is suspended over source substrate **10** and is physically attached to source substrate **10** only by tether **40** connected to anchor portion **16** of source substrate **10**. Micro-device **20** can then be micro-transfer printed (e.g., with an elastomeric stamp) from source substrate **10** to a target substrate **60**, as illustrated in FIG. **4**, discussed subsequently.

(35) Where etch hole **30** and micro-device **20** are both substantially rectangular, etch hole **30** has an etch-hole edge **32**, and micro-device **20** has a micro-device edge **22**, etch-hole edge **32** can be substantially parallel to micro-device edge **22** (as shown in FIGS. **1**A-**2**B). Etch hole **30** can have an etch-hole edge **32** with an etch-hole width W.sub.E in the range of 5 microns to 20 microns. According to some embodiments, micro-device **20** has a micro-device edge **22** in the range of 10 microns to 5 mm. For example, micro-device **20** can have a micro-device length L.sub.M no greater than 5 mm, 2 mm, 1 mm, 500 microns, 200 microns, 150 microns, or 100 microns.

(36) Embodiments of the present disclosure can enable an efficient use of source substrate **10** area, reducing micro-device **20** manufacturing costs. According to some embodiments and as illustrated in the plan view of FIG. **3**A and partial perspective of FIG. **3**B, anchor portion **16** can be a central portion to which is physically attached multiple tethers **40** physically connecting corresponding micro-devices **20** with etch holes **30** over sacrificial portions **14**. Each micro-device **20** and corresponding tether **40** physically connecting micro-device **20** to anchor portion **16** is a unit that is rotated with respect to all other units physically connected with tethers **40** to the same, common anchor portion **16**. Since such a configuration and source substrate **10** layout can have smaller anchor portions **16**, the arrangement illustrated in FIG. **3**A can more efficiently use the area of source substrate **10**, reducing manufacturing costs. That is, more of the area of source substrate **10** is devoted to micro-devices **20**. Etch holes **30** enable a faster and more efficient release of micro-devices **20** in the configuration illustrated in FIGS. **3**A and **3**B from source substrate **10** for sacrificial portions **14** comprising an anisotropically etchable material, such as silicon {100}. Thus, according to embodiments of the present disclosure, a micro-device structure **92** comprises a source substrate **10** having a sacrificial layer **12** comprising sacrificial portions **14** adjacent to an anchor portion **16**, micro-devices **20** disposed completely over each sacrificial portion **14**, and a tether **40** that physically connects each micro-device **20** to anchor portion **16**. Each micro-device **20** and corresponding physically connected tether **40** are rotated with respect to each other micro-device

20 and corresponding physically connected tether **40**. In some embodiments, three micro-devices **20** are each physically connected to a common anchor portion **16** with a different tether **40**. In some embodiments and as shown in FIGS. **3**A and **3**B, four micro-devices **20** are each physically connected to a common anchor portion **16** with a different tether **40** and are each rotated by 90 degrees from a neighboring micro-device **20**. In some embodiments, micro-devices **20** and corresponding tether **40** are reflections of other micro-devices **20** and tether **40** (not shown in the Figures). In some embodiments, micro-devices **20** are square (e.g., have an aspect ratio of 1:1) and are not reflected or rotated but are physically connected by tethers **40** at different corners of micro-device **20** to anchor portion **60**. As used in this context, at different corners also includes configurations in which tether **40** is physically connected to micro-device **20** closer to a corner of micro-device **20** than to a center or to another corner along the same edge of micro-device **20**. Physically connecting tethers **40** to micro-devices **20** at the corners of micro-devices **20** as shown in FIGS. **3**A and **3**B can provide a denser configuration and better utilization of source substrate **10** surface area, but embodiments of the present disclosure are not limited to such configurations.

(37) According to some embodiments and as shown in FIG. **4**, a micro-device structure **94** (e.g., a micro-transfer printed micro-device structure **92**) comprises a target substrate **60** and a micro-device **20** (or multiple micro-devices **20**) disposed on or over target substrate **60**. Micro-device **20** can have a top side **24** and a bottom side **26** opposite top side **24**. Bottom side **26** can be adjacent to target substrate **60** and top side **24** is on a side of micro-device **20** opposite target substrate **60**. Each micro-device **20** can comprise an etch hole **30** that extends through micro-device **20** from top side **24** to bottom side **26**. Etch hole **30** can expose target substrate **60** or a layer on target substrate **60** (e.g., an adhesive layer). Etch hole **30** can have tapered side walls so that a top side (top opening) of etch hole **30** has a greater area than a bottom side (bottom opening) of etch hole **30** adjacent to target substrate **60**. A broken (e.g., fractured) or separated tether **40** can be physically connected to micro-device **20**, for example as a consequence of micro-transfer printing micro-device **20** from source substrate **10** to target substrate **60**. Etch hole **30** can have an etch-hole edge **32** with an etch-hole width W.sub.E in the range of 5 microns to 20 microns. Micro-device **20** can have a micro-device length L.sub.M no greater than 5 mm, 2 mm, 1 mm, 500 microns, 200 microns, 150 microns, or 100 microns. Micro-devices **20** on or over target substrate **60** can be coated with or encapsulated by, an encapsulating layer, for example a resin, planarizing material, silicon dioxide, or silicon nitride.

(38) According to some embodiments and as illustrated in the flow diagram of FIG. **5**, a method of making a micro-device structure **90** comprises providing a source substrate **10** comprising a sacrificial layer **12** comprising a sacrificial portion **14** adjacent to an anchor portion **16** in step **100** and providing a micro-device **20** disposed completely over sacrificial portion **14** in step **110**. Micro-device **20** can have a top side **24** opposite sacrificial layer **12** (or sacrificial portion **14**) and a bottom side **26** adjacent to sacrificial layer **12** (or sacrificial portion **14**) and can comprise etch hole **30** that extends through micro-device **20** from top side **24** to bottom side **26**, exposing a portion of sacrificial portion **14**. Tether **40** physically connects micro-device **20** to anchor portion **16**. An etchant is provided in step **120**, and sacrificial portion **14** is etched in step **130**. At least the portion of sacrificial portion **14** exposed by etch hole **30** is etched by the etchant passing through etch hole **30**, thereby forming gap **15** between micro-device **20** and source substrate **10**, suspending micro-device **20** over source substrate **10**, and forming a micro-device structure **90** as shown in FIGS. **2**A and **2**B. By forming gap **15** and suspending micro-device **20** over source substrate **10**, micro-device **20** is rendered micro-transfer printable from source substrate **10**. By etching through etch hole **30**, sacrificial portion **14** material directly beneath micro-device **20** is etched and, accelerates removal of sacrificial portion **14** material (e.g., where slow-etching crystal planes of an anisotropic source substrate **10** (and sacrificial portion **14**) are exposed) and reduces any damage to micro-device **20**, tether **40**, and anchor portion **16**.

(39) According to some embodiments and as illustrated in FIG. **4** and with reference further to FIG.

5, methods of the present disclosure can further comprise providing a stamp in step **140** and target substrate **60** in step **170**, contacting micro-device **20** with the stamp to adhere micro-device **20** to the stamp in step **150**, removing the stamp from source substrate **10** in step **160**, thereby breaking (e.g., fracturing) or separating tether **40**, pressing micro-device **20** to target substrate **60** in step **180** to adhere micro-device **20** to target substrate **60**, and removing the stamp in step **190**, forming a micro-device structure **92** as shown in FIG. **4**.

(40) In some embodiments of methods of the present disclosure, sacrificial portions **14** can be etched via wet etch, for example when exposed to a hot bath of tetramethylammonium hydroxide (TMAH) or potassium hydroxide (KOH) or any appropriate chemistry or a gaseous etch.

(41) Embodiments of the present disclosure have been modeled and their etching characteristics analyzed for tether widths W.sub.T varying from 6 to 16 microns, etch-hole width W.sub.E varying from 9-18 microns with various aspect ratios, anchor portion **16** having a polygonal cross section and diameters of 30-80 microns, and micro-devices **20** having micro-device lengths L.sub.M of 150 microns at various angles of micro-device edge **22** to source substrate flat **18** ranging from 0 to 90 degrees (e.g., including 0 degrees, 22.5 degrees, 45 degrees, 67.5 degrees, and 90 degrees). As shown in the graph of FIG. **6**, etch holes **30** with an etch-hole width W.sub.E that is greater than a tether width W.sub.T of a tether **40** successfully release in an acceptable time period. The points graphed in FIG. **6** represent a simulated examples of embodiments that have a particular etch-hole width W.sub.E and tether width W.sub.T that were simulated to release in a single, same acceptable release time, each point representing a different combination of tether width W.sub.T and etch-hole width W.sub.E. For example, a tether width W.sub.T of 8 μm and an etch-hole width W.sub.E of 10 μm can be used to release a micro-device **20** in an acceptable release time. Likewise, a tether width W.sub.T of 10 μm and an etch-hole width W.sub.E of 12 μm can be used to release a micro-device **20** in the same acceptable release time, as can a tether width W.sub.T of 12 μm and an etch-hole width W.sub.E of 14 μm. As can be seen from the simulated data plotted in FIG. **6**, micro-device **20** can be released (e.g., to be micro-transfer printed) by etching for the chosen preferable acceptable release time when etch-hole width W.sub.E is greater than tether width W.sub.T. In some embodiments, less desirable etch times are required for different combinations of tether widths W.sub.T and etch-hole widths W.sub.E. An acceptable release time can be chosen based on, for example, commercial considerations (e.g., throughput) and/or a desired extent of release of (e.g., amount of etching under) micro-device **20** to be made (e.g., based on operational parameters of a subsequent micro-transfer printing to be performed).

(42) Micro-device **20** can be or can include, for example, any one or more of an electronic component, a piezoelectric device, an integrated circuit, an electromechanical filter, an acoustic resonator, an antenna, a micro-heater, a micro-fluidic structure for containing and constraining fluids, a micro-mechanical device, and a power source, for example a piezo-electric power source. In some embodiments of the present disclosure and as illustrated in FIG. **7**, micro-device **20** is or comprises an antenna **70**. Antenna **70** can be disposed around a periphery of micro-device **20** and can be a spiral antenna, can have a height greater than a width over a surface or substrate of micro-device **20**. In some embodiments of the present disclosure and as illustrated in FIG. **8**, micro-device **20** is or comprises a micro-heater **72**, for example a strip micro-heater **72** comprising a resistive electrical conductor. In some embodiments of the present disclosure and as illustrated in FIG. **9**, micro-device **20** is or comprises a micro- or nano-fluidic structure **74**, for example comprising one or more etch holes **30** and one or more channels **78** for containing liquids that can flow or wick between etch holes **30** through channel(s) **78**. A micro- or nano-fluidic structure **74** can further include one or more pumps, valves, or other fluid control elements. A micro- or nano-fluidic structure **74** can be an open surface structure or be enclosed. Etch holes **30** can act as an inlet or outlet for a micro- or nano-fluidic structure in addition to facilitating etching of sacrificial portion **14** material during micro-device **20** release. In some embodiments of the present disclosure and as illustrated in FIG. **10**, micro-device **20** is or comprises a power device **76**, for example a piezo-

electric device with one or more beams (or cantilevers) that vibrate in response to mechanical stimulation and generate electrical power. More generally, embodiments of the present disclosure can be applied to micro-devices **20** that have space for etch hole **30**, for example due to their size, shape, or layout.

(43) In some embodiments and as illustrated in FIGS. **11**A-B, source substrate **10** is an intermediate structure **95** wafer. Source substrate **10** has sacrificial layer **12** comprising sacrificial portion **14**. Intermediate substrate **52** is disposed completely over sacrificial portion **14**. Intermediate substrate **52** has an etch hole **30** extending through intermediate substrate **52** from top side **24** to bottom side **26**. One or more micro-devices **20** are disposed on intermediate substrate **52**, for example on top side **24** thereof as shown. Micro-device(s) **20** can be non-native to intermediate substrate **52**. Micro-device(s) **20** can be electronic, optical, or optoelectronic devices that can be electrically, optically, or both electrically and optically interconnected on intermediate substrate **52** (e.g., by wire(s) or trace(s) or light pipe(s)). One of micro-device(s) **20** can itself include an etch hole **30** that is disposed in alignment with etch hole **30** of intermediate substrate **52** such that etchant can flow through both. Given that intermediate substrate **52** is relatively large (e.g., at least 2×, at least 4×, or at least 10× larger than micro-device **20**), etch hole **30** can assist in promoting fast and complete release of intermediate structure **95** from source substrate **10** such that intermediate structure **95** can be printed to target substrate **60**. Intermediate structure **95** is connected to anchor portion **16** with tether **40** to maintain orientation and alignment of intermediate structure **95** after release. Tether **40** can be broken or separated to print intermediate structure **95**. Methods of forming intermediate structures **95** that can be readily adapted to include etch holes **30** in intermediate substrates **52** are disclosed in U.S. Pat. No. 10,468,398, the disclosure of which is hereby incorporated by reference.

(44) Although many figures presented herein often illustrate a single micro-device or intermediate substrate in a wafer for simplicity, one of ordinary skill in the art will appreciate that there will generally be many such micro-devices or such intermediate substrates disposed on a wafer (e.g., in a two-dimensional array).

(45) Etch hole **30** can have an area relative to an area of micro-device **20** that is no greater than 10%, no greater than 5%, no greater than 1%, no greater than 0.5%, or no greater than 0.1%. For example, a micro-device **20** that has a micro-device length L.sub.M and micro-device width W.sub.M of 100 microns each and a square etch hole **30** of 10 microns on each etch-hole side, will have an etch hole area of 1%. For the simulated micro-device **20** having a micro-device length L.sub.M and micro-device width W.sub.M of 150 microns each and a square etch hole **30** of 10 microns on each etch-hole side, will have an etch hole area of 0.44%.

(46) In some embodiments of the present disclosure, the etching process to remove sacrificial portions **14** of sacrificial layers **12** underneath releasable micro-devices **20** has a crystallographic dependence, etching faster in some directions of a crystal structure and slower in other directions of the crystal structure. Corner structures etch at different rates because of differences in the number of available dangling bonds that are susceptible to different etch rates. For example, in certain crystal structures and crystallographic orientations (e.g., with cubic symmetry and {100} orientation), an atom that is normally connected to four neighbors (e.g., in the bulk) may only be connected to two neighbors at a convex corner but would be connected to three neighbors at a concave corner due to the respective surface geometries of such corners. In certain crystal structures, a crystal atom normally connected to eight neighbors may only be connected to three neighbors at a convex corner but would be connected to seven neighbors at a concave corner. Therefore, convex or exterior corners of a crystalline structure(s) in sacrificial layer **12** etch relatively quickly, progressively etching and producing etch fronts to match the fast etching planes in the crystal, due to lower coordination (increased dangling bonds) for exposed atoms. Concave or internal corners of a crystalline structure(s) in sacrificial layer **12** have fewer susceptible dangling bonds and therefore etch more (relatively) slowly, forming a slowly moving or pinned/stopped etch front defined by the slowly etching planes (in many cases at a rate that is hardly measurable).

Resulting etch fronts of the etchants can form and maintain a local shape characterized by the internal/concave corners. Some release layers (also referred to as a sacrificial layer **12**) that exhibit this kind of crystallographic selectivity include Si {1 1 1}, Si {1 0 0}, InAlP, InP, GaAs, InGaAs, AlGaAs, GaSb, GaAlSb, AlSb, InSb, InGaAlSbAs, InAlSb, and InGaP.

(47) Therefore, according to some embodiments of the present disclosure, the location, size, and shape of etch holes **30** can be chosen with regard to concave corners of micro-device **20** to reduce or eliminate formation of pinned etch front(s), which can inhibit micro-device **20** release. The effect of pinned etch front(s) can be even more significant when micro-devices **20** are large (e.g., no less than 100 μm in at least one dimension).

(48) As shown in FIGS. **1**B and **1**D-**3**B, micro-devices **20** can be attached to anchor portion **16** (anchor **16**) by tether **40** (e.g., a single tether **40**). In order to enable effective pick-up (e.g., by a stamp for micro-transfer printing), tether **40** has a tether width W.sub.T along an edge of micro-device **20** that is less than a length of the edge of micro-device **20** (e.g., as shown in FIG. **1**D). Tether **40** therefore forms a concave corner with micro-device **20**, inhibiting etching under micro-device **20** and under tether **40**. According to embodiments of the present disclosure, micro-device **20** comprises or has a concave corner if tether **40** contacts (e.g., touches or intersects) micro-device **20** and tether **40** has a width W.sub.T along an edge of micro-device **20** that is less than the length of the edge of micro-device **20**. Furthermore, if multiple tethers **40** physically connect micro-device **20** to anchor portion **16**, multiple concave corners are present.

(49) In some embodiments in which micro-devices **20** are not intended for transfer printing, tether **40** can serve as a cantilever attached to micro-device **20**, for example if tether **40**, micro-device **20**, or both, form a piezoelectric device and micro-device **20** is designed to oscillate vertically with respect to a surface of source substrate **10** (source wafer **10**). In some such embodiments, tethers **40** can also form concave corners with micro-device **20**. Likewise, in some embodiments in which tethers **40** are intended to etch entirely away so that micro-devices **20** fall into gap **15** or fall away from source substrate **10** (e.g., for subsequent transfer and/or collection), tethers **40** form concave corners with micro-devices **20**.

(50) Embodiments of the present disclosure illustrated in FIGS. **1**A-**3**B, **7**, **8**, and **10** illustrate micro-devices **20** with a single etch hole **30** and a single tether **40**. FIG. **12** illustrates micro-device structure **90** with micro-device **20** attached to anchor portion **16** with two tethers **40**. FIG. **13** illustrates micro-device structure **90** with micro-device **20** attached to anchor portion **16** with two tethers **40** on each edge side **22** of rectangular micro-device **20**. Multiple tethers **40** can prevent micro-device **20** from contacting source substrate **10**, even though micro-device **20** is suspended over gap **15** by a distance D, especially where micro-device **20** is relatively large or has a large aspect ratio (e.g., forms a thin rectangle). In both FIG. **12** and FIG. **13**, two etch holes **30** extend through micro-device **20** from top side **24** of micro-device **20** to bottom side **26** of micro-device **20** (not shown in FIGS. **12** and **13**, see FIG. **11**A). Bottom side **26** is adjacent to and exposes source substrate **10** or a sacrificial portion **14** (e.g., as shown in FIGS. **11**A and **11**B) or gap **15** (e.g., as shown in FIGS. **12** and **13**).

(51) According to some embodiments of the present disclosure, micro-devices **20** can have etch holes **30** with a large aspect ratio, for example no less than 2:1, 3:1, 5:1, 10:1, 20:1, or even 50:1. Such embodiments of high-aspect-ratio etch holes **30** can provide ingress to etchants attacking fast-etching crystal planes in areas that would otherwise etch last or would cause undesired long etch processes. Larger aspect ratios can expose greater fast-etch plane area for an equivalent etch hole **30** size (e.g., to balance etch rate and usable micro-device **20** area). Etching directly underneath relatively large micro-devices **20**, for example micro-devices **20** that are no less than 100 microns in length or width (e.g., both) (e.g., no less than 200 microns in length or width, no less than 500 microns in length or width, no less than 750 microns in length or width, no less than 1 mm in length or width, or even larger in length or width) can then be accelerated or timed accordingly to ensure structural support throughout the release process. High-aspect-ratio etch holes **30** can be

oriented with respect to the crystallographic structure and orientation of source wafer **10**, e.g., as illustrated and discussed with respect to FIG. **1**A. Etch holes **30** with a high (large) aspect ratio (e.g., a large ratio of length to width where length is greater than width) can expose a greater portion of a desired etch plane while reducing the total area of etch holes **30** (e.g., to maintain the structural integrity of micro-device **20**). In some embodiments, mass is added to micro-device **20** to compensate for the loss of mass in micro-device **20** from etch holes **30**.

(52) FIGS. **14** and **15** illustrate micro-device structures **90** with micro-devices **20** and four high-aspect-ratio etch holes **30**. In FIGS. **14** and **15**A-**15**C, etch holes **30** can each comprise two crossing line segments forming extended arms of a common single etch hole **30** that each have a much greater length than width. As used herein, line segments are high-aspect-ratio rectangles where length is the longer dimension and width is the shorter dimension of each line segment in a direction parallel to top side **24**. As used herein, a single etch hole **30** comprising two or more connected line segments is considered a high-aspect-ratio etch hole **30** because each of the two or more connected line segments have a large aspect ratio. Thus, according to some embodiments of the present disclosure, a micro-device structure **90** comprises a micro-device **20** having a top side **24** and a bottom side **26** and one or more etch holes **30**. Each of the one or more etch holes **30** is an etch hole **30** that extends through micro-device **20** from top side **24** to bottom side **26** so that, for example, an etchant can flow through the etch holes **30**. One or more tethers **40** physically connect micro-device **20** to anchor portion **16**. At least one etch hole **30** has an aspect ratio no less than 2:1.

(53) According to some embodiments, micro-device **20** can be disposed over and native to source substrate **10**. Source substrate **10** can comprise a sacrificial layer **12** comprising anchor portions **16** laterally separated by sacrificial portions **14** in a direction parallel to a surface of source substrate **10** or top side **24**, as shown in FIGS. **11**A and **11**B. Anchor portions **16** can be a part of source substrate **10** or a structure disposed on source substrate **10**. According to some embodiments, micro-device structure **90** is disposed on a target substrate **60** or intermediate substrate **52** (e.g., as shown in FIG. **4**, for example by micro-transfer printing). Target substrate **60** or intermediate substrate **52** can comprise a gap **15** (or cavity **15**) beneath micro-device **20**. In some embodiments, gap **15** and tether **40** enable micro-transfer printing micro-device **20** from source wafer **10**. In some embodiments, gap **15** between micro-device **20** and source, target, or intermediate substrate **10**, **60**, **52** provides room for micro-device **20** to physically move, for example oscillate or vibrate in a direction orthogonal to top side **24**, in embodiments in which tether **40** provides a cantilever for micro-device **20**, for example if tether **40** or micro-device **20**, or both, comprise a piezoelectric device comprising piezoelectric material, e.g., an acoustic filter or power generator. In some such embodiments, micro-device **20** can be or include a mass at the distal end of a cantilever (tether **40**) that mechanically stresses the piezoelectric material to generate electrical power when mechanically perturbed (e.g., by shaking or otherwise accelerating micro-device **20**).

(54) As will be appreciated by those knowledgeable in crystallography and etching processes, a wide variety of etch holes **30**, especially high-aspect-ratio etch holes **30**, can be disposed in micro-device **20**. FIGS. **14** to **23** illustrate a variety of etch hole **30** arrangements with respect to a variety of tethers **40**, for example a single tether **40** on a single edge side **22** of micro-device **20** (e.g., as shown in FIGS. **22** and **23**), two or more tethers **40** (e.g., as shown in FIGS. **12**-**21**C), two or more tethers **40** on a single edge side **22** of micro-device **20** (e.g., as shown in FIGS. **12** and **16**-**18**), one tether **40** per micro-device **20** edge side **22** (e.g., as shown in FIGS. **14** and **15**), or two or more tethers **40** per micro-device **20** edge side **22** (e.g., as shown in FIGS. **13**, **19**-**21**C).

(55) According to some embodiments of the present disclosure, etch holes **30** can be disposed in micro-device **20** so as to reduce the time required to under-etch micro-device **20** and completely release it from its native source wafer **10**. Locations at which an etch front can be pinned (e.g., stopped), hard to reach by wet chemicals, or greatly slowed compared to etching along fast-etching planes, are illustrated as pinned etch fronts **34** (pinned etch planes **34**) with dashed straight lines or rectangular boxes in FIGS. **15** and **21**B and extend from concave corners **36**. Etch holes **30** are

disposed to contact (e.g., touch or intersect) or overlap pinned etch planes **34** so that etchant can attack pinned etch planes **34** from a different fast-etching plane direction and thus etch otherwise pinned etch planes **34**. FIGS. **15**A and **21**B illustrate the overlap of etch holes **30** and pinned etch planes **34** associated with each individual etch hole **30**. Because pinned etch planes **34** associated with etch holes **30** overlap, each pinned etch plane **30** can be etched from a different non-pinned (e.g., fast etching) plane, resulting in complete under-etching (release) of micro-device **20** (e.g., within a commercially practical time frame, such as two hours, one hour, 30 minutes, or 15 minutes). In FIG. **21**B, the pinned etch planes **34** resulting from concave corners **36** at the juncture of tethers **40** and micro-device **20** are not marked but are interrupted by diagonal etch holes **30** at 45 degrees to a micro-device **20** edge side **22** and the extended arm of plus-shaped etch holes **30** extending orthogonally to edge side **22** of micro-device **20**.

(56) Etch holes **30** can expose fast-etching planes at otherwise pinned etch plane **34** locations and can also be disposed in micro-device **20** to increase the etch rate and decrease the etch time even if no pinned etch planes **34** are present. In some cases, no or few concave corners **36** are present, but micro-device **20** is sufficiently large that is desirable to accelerate the release process to avoid damage to micro-device **20**, tether **40**, or anchor portion **16** since, if the wet etch rates are too high, the etchant can also damage micro-device **20**, tether **40**, or anchor portion **16** as the etching selectivity between the sacrificial layer and other materials is generally not infinite.

(57) According to some embodiments of the present disclosure, etch holes **30** can be rectangular (e.g., as shown in FIGS. **12** and **13**) and can have a large aspect ratio, for example a length to width ratio of no less than 2:1, no less than 4:1, or no less than 8:1. An etch hole **30** can comprise connected line segments, where each line segment is a rectangle with a large aspect ratio, for example forming plus (‘+’) sign shapes, X ‘(X’) shapes, cross shapes, Y (‘Y’) shapes, and T (‘T’) shapes. A combination of connected line segments or other rectangular or curved line segments forms a single etch hole **30**. In some embodiments, etch hole **30** can be disposed near or substantially at or intersecting a geometric center of micro-device **20**.

(58) Etch holes **30** can also be disposed in micro-device **20** so as to progressively etch sacrificial portion **14** in a desired temporal progression. Etching sacrificial portion **14** to form gap **15** can result in etch byproducts such as gases that form bubbles that mechanically perturb micro-device **20** and tether **40**, possibly undesirably cracking tether **40** or micro-device **20**. Thus, it can be helpful, in some embodiments of the present disclosure, to progressively etch sacrificial portion **14** so that micro-device **20** and tether **40** are supported (not completely undercut) until most of the etching process is complete, thus providing mechanical support for micro-device **20** and tether **40** during most of the etching process. Once most of the etching process is complete, the remainder of the etch can be relatively small and therefor relatively gentle, thus preserving the structural integrity of micro-device **20** and tether **40** during the under-etching process.

(59) In order to ensure that micro-devices **20** are released despite any pinned etch fronts resulting from any concave corners **36** of micro-device **20** and according to embodiments of the present disclosure and as illustrated in FIG. **15**B, etch holes **30** are disposed in micro-device **20** so that any straight line **37** parallel to an edge side **22** of micro-device **20** drawn from a concave corner **36** on a first edge side **22** of micro-device **20** to a second edge side **22** of micro-device **20** opposite the first edge side **22** contacts (e.g., touches or intersects) an etch hole **30**. As used herein with respect to etch hole **30** and micro-device **20** geometry, “a side” (or “any side”) of micro-device **20** (without reference to its particular location) is also an edge side **22** of micro-device **20** and does not include top side **24** or bottom side **26** of micro-device **20** (which are specifically referred to by their location—top and bottom, respectively). This requirement is particularly (but not exclusively) useful where tethers **40** are attached to micro-device **20** on a single edge side **22** of micro-device **20** (e.g., as shown in FIGS. **12**, **14-18**, and **22-23**. Micro-devices **20** can have a polygonal perimeter (e.g., a rectangular perimeter) and edge sides **22** of micro-device **20** can have a shape of a line segment that is a portion of the polygonal perimeter.

(60) Similarly, in order to ensure that micro-devices **20** are released despite any pinned etch fronts **34** resulting from any concave corners **36** of micro-device **20** and according to embodiments of the present disclosure and as illustrated in FIG. **15**C, etch holes **30** are disposed in micro-device **20** so that any orthogonal line segment angle **38** drawn from a first concave corner **36** on a first edge side **22** of micro-device **20** to a second concave corner **36** on a second edge side **22** of micro-device **20** adjacent to the first edge side **22** contacts (e.g., touches or intersects) an etch hole **30**. Adjacent edge sides **22** of micro-device **20** are edges side **22** that contact each other, e.g., at a corner (whether concave or convex). An orthogonal line segment angle **38** comprises two line segments that contact each other at an end of each line segment and that extend orthogonally (at right angles) to each other, thus forming a corner, e.g., two adjacent sides of a square or rectangle. This requirement can be particularly (but not exclusively) useful where tethers **40** are attached to micro-device **20** on each edge side **22** of micro-device **20** (e.g., as shown in FIGS. **13**, **14-15**, and **19-21**C.
(61) According to embodiments of the present disclosure, an etch hole **30** has two or more portions (e.g., line segments or high-aspect-ratio line segments) that extend in different directions (e.g., orthogonally forming plus, X, Y, cross, or T shapes) and is disposed closer to a center of micro-device **20** than to any edge side **22** of micro-device **20**. In some embodiments, etch hole **30** intersects or is disposed at a center of micro-device **20**, e.g., as shown in FIGS. **22-23** and also FIGS. **15-17**, **19-21**C. In some such embodiments, etch hole **30** can be the only etch hole **30** in micro-device **20** (e.g., a singular etch hole **30**) as shown in FIGS. **22** and **23**.
(62) Etch holes **30** can be disposed in such a way that they do not cut micro-device **20** into physically separate portions, e.g., micro-device **20** is a unitary and contiguous micro-device **20** despite etch holes **30** disposed through micro-device **20**. According to some embodiments, one or more etch holes **30** extend to an edge of micro-device **20** (e.g., intersect an edge of micro-device **20** as shown in FIGS. **15**, **17**, **19**, **21**A-**21**C). According to some embodiments, etch holes **30** do not extend to an edge of micro-device **20** (e.g., as shown in FIGS. **12-15**, **19**, **20**, **22**, **23**). In some embodiments, at least one etch hole **30** is disposed at an angle that is not parallel or orthogonal to an edge side **22** of micro-device **20** (for example disposed on a diagonal such as a 45 degree diagonal with respect to edge sides **22** of micro-device **20** as shown in FIGS. **16**, **19-21**C).
(63) In some embodiments, anchor portion **16** surrounds micro-device **20** in a horizontal direction (e.g., parallel with or in a plane defined by top side **24** of micro-device **20** or by a surface of source substrate **10** or target substrate **60**) as shown in the plan views of FIGS. **12-15**, **22-23** and cross section of FIG. **24**.
(64) According to some embodiments of the present disclosure, micro-device structure **90** comprises a source wafer **10** (e.g., a native source wafer **10**) and anchor portion **16** is disposed on or is a portion of source wafer **10** and micro-device **20** is separated from source wafer **10** by cavity **15** (gap **15**) or sacrificial portion **14**. Source wafer **10** can have a crystalline structure that is anisotropically etchable. Source substrate can comprise silicon, e.g., silicon {100} or silicon {111}. In some embodiments, source substrate **10** has a crystalline structure with a {100} orientation and bottom side **26** of micro-device **20** is substantially parallel to a {100} crystal plane at a surface of source substrate **10**. In some embodiments, micro-device **20** has a micro-device edge direction oriented at an angle from 30 to 60 degrees with respect to a {110} crystal plane of the crystalline structure. In some embodiments, micro-device **20** comprises a micro-device edge having a direction oriented at an angle of substantially 45 degrees with respect to the {110} crystal plane. In some embodiments, source substrate **10** has a crystalline structure with a {100} crystal plane, micro-device **20** has a micro-device length greater than a micro-device width and the micro-device length is in a direction that is substantially parallel to the {100} crystal plane.
(65) According to some embodiments of the present disclosure, micro-device structure **90** comprises a micro-device **20** and one or more etch holes **30**, micro-device **20** and at least one etch hole **30** are both rectangular, etch hole **30** has an etch-hole edge, micro-device **20** has a micro-device edge, and the etch-hole edge is substantially parallel, substantially orthogonal, or

substantially at 45 degrees to the micro-device edge. By orienting the etch-hole edge at such angles to the micro-device edge and orienting the micro-device edge with respect to the native source substrate **10** crystal structure, fast etch planes in native source substrate **10** can be exposed by etch-holes **30**. According to embodiments of the present disclosure, etch hole edges or line segments are disposed substantially parallel, substantially perpendicular, substantially at 30 degrees, or substantially at 40 degrees with respect to the wafer flat. By substantially is meant as designed or within manufacturing limitations.

(66) According to some embodiments of the present disclosure and as illustrated in FIG. **24**, micro-device structure **90** can be micro-transfer printed from native source substrate **10** to a target or intermediate substrate **60**, **52**. In some such embodiments, a micro-device system **94** comprises a target or intermediate substrate **60**, **52**, one or more micro-device structures **90** disposed on target or intermediate substrate **60**, **52**, and a broken (e.g., fractured) or separated structure tether **42** physically attached to anchor portion **16**. FIGS. **14** and **15** illustrate structure tethers **42** attached to anchor portions **16**. Such embodiments can integrate micro-device structures **90** into electronic micro-device systems **94** for a variety of applications.

(67) According to some embodiments of the present disclosure and as illustrated in the cross sections of FIGS. **25**A and **25**B and the plan view and detail of FIG. **25**C, etch holes **30** in micro-devices **20** can be coated with an etch-hole protective coating **33** that is relatively impervious to an etchant that etches sacrificial portion **14**, e.g., as shown in FIGS. **25**A and **25**C. Such an etch-hole protective coating can also be provided on the top or bottom of micro-device **20** (as shown in FIG. **25**B) to protect micro-device **20** from the sacrificial portion **14** etchant. FIGS. **25**A and **25**B illustrate micro-device **20** with two etch holes **30** in micro-device **20** and FIG. **25**C illustrates micro-device **20** with one etch hole **30** in micro-device **20**. Etch-hole protective coating **33** can comprise a variety of materials depending on the specific materials and chemistries of micro-device **20**, sacrificial portion **14**, source substrate **10**, or intermediate substrate **52**. In some embodiments, micro-devices **20** can comprise a same material as sacrificial portions **14**, for example silicon, and etch-hole protective coating **33** can comprise one or more oxides or nitrides, for example silicon dioxide or silicon nitride, which are especially useful for a TMAH etchant, as an example. In some embodiments, other materials can be disposed over a protective layer on a silicon substrate and the protective layer materials can be etch-hole protective coatings **33**. For example, in some embodiments, micro-devices **20** can comprise potassium sodium niobate (KNN) disposed over a silicon substrate and etch-hole protective coating **33** can comprise one or more oxides or nitrides, for example silicon dioxide or silicon nitride, which are especially useful for a TMAH etchant, as an example. In some embodiments using an SOI (semiconductor on insulator) wafer with a buried oxide insulating layer and a non-silicon semiconductor, a layer of amorphous silicon can provide an effective etch-hole protective coating **33** against an etchant such as concentrated hydrofluoric acid. Etch-hole protective coating **33** can be disposed by evaporation or sputtering after etch holes **30** are formed and can have a thickness of a few microns or less than a micron.

(68) Embodiments of the present disclosure encompass a wide variety of etch hole **30** sizes, shapes, and arrangements useful for a corresponding variety of micro-devices **20** and applications. According to some embodiments, a first etch hole **30** extends into micro-device **20** from a first side, a second etch hole **30** extends into micro-device **20** from a second side that is opposite the first side, and a third etch hole **30** extends into micro-device **20** from a third side different from the first side and from the second side. Third etch hole **30** can comprise a first portion that intersects the third side and a second portion that intersects an end of the first portion at a point other than an end of the second portion. In some embodiments, the first portion bisects the second portion. In some embodiments, first etch hole **30** and second etch hole **30** are parallel. In some embodiments, first etch hole **30** and second etch hole **30** are collinear. In some embodiments, the first portion extends into micro-device **20** past first etch hole **30** and second etch hole **30**. Such etch hole **30** arrangements can facilitate etching where concave corners **36** are present in micro-device **20**.

(69) According to some embodiments, a plurality of etch holes **30** are disposed in a symmetric arrangement, the symmetric arrangement having a multifold symmetry and comprising a unit cell comprising a first portion of a first etch hole **30** extending outward from a center of micro-device **20**, a second etch hole **30** that is an interior etch hole **30** and disposed parallel to a side of micro-device **20**, and a third etch hole **30** that is an interior etch hole **30** collinear with a center of micro-device **20**. Interior etch holes **30** do not contact an edge or side of micro-device **20**. In some embodiments, the symmetric arrangement has a two-fold symmetry or a four-fold symmetry (e.g., as shown in FIGS. **19**, **20**, and **21**A-C). In some embodiments, first etch hole **30** has a cross shape. In some embodiments, second etch hole **30** and third etch hole **30** are straight etch holes **30**, e.g., rectangular. In some embodiments, third etch hole **30** is collinear with both a center of micro-device **20** and a corner of micro-device **20**. Symmetric etch hole **30** arrangements can reduce stress during etching.
(70) According to some embodiments, a first etch hole **30** comprises a first portion and a second portion (e.g., first and second line segments). The first portion and the second portion intersect at respective interior points of micro-device **20** and a center of first etch hole **30** is disposed closer to a center of micro-device **20** than any side of micro-device **20**. According to some embodiments, the first portion and the second portion each intersect the center of micro-device **20**. According to some embodiments, the first portion and the second portion form a cross. According to some embodiments, the first portion has a length equal to a length of the second portion. According to some embodiments, first etch hole **30** is an only etch hole **30**.
(71) According to some embodiments, micro-device system **94** comprises a fractured structure tether **42**. According to some embodiments, micro-device **20** has a thickness less than 1 mm (e.g., no greater than 500, 200, 100, 50, 20, 10, 5, 1, or 0.5 microns). According to some embodiments, micro-device **20** has a length or width less than 1 mm (e.g., no greater than 500, 200, 100, 50, 20, or 10 microns).
(72) According to some embodiments of the present disclosure, a micro-device system **94** comprises a micro-device **20** disposed on a target substrate **60** (or intermediate substrate **52**), wherein micro-device **20** is non-native to target substrate **60**. Micro-device structure **90** can have a thickness no greater than 1 mm (e.g., no greater than 500, 200, 100, 50, 20, or 10 microns). Micro-device structure **90** can have a length or width no greater than 1 mm (e.g., no greater than 500, 200, 100, 50, 20, or 10 microns).
(73) Micro-device **20** can be native to source substrate **10**, or non-native to source substrate **10**. A micro-device **20** can be any device that has at least one dimension that is in the micron range, for example having a planar extent from 2 microns by 5 microns to 200 microns by 500 microns (e.g., an extent of 2 microns by 5 microns, 20 microns by 50 microns, or 200 microns by 500 microns) and, optionally, a thickness of from 200 nm to 200 microns (e.g., at least or no more than 2 microns, 20 microns, or 200 microns). In some embodiments, micro-device **20** has a dimension as large as, or larger than 5 mm. Micro-device **20** can have any suitable aspect ratio or size in any dimension and any useful shape, for example a rectangular cross section.
(74) In some embodiments of the present disclosure, micro-devices **20** are small integrated circuits, which may be referred to as chiplets, having a thin substrate with at least one of (i) a thickness of only a few microns, for example less than or equal to 25 microns, less than or equal to 15 microns, or less than or equal to 10 microns, (ii) a width of 5-1000 microns (e.g., 5-10 microns, 10-50 microns, 50-100 microns, or 100-1000 microns), and (iii) a length of 5-1000 microns (e.g., 5-10 microns, 10-50 microns, 50-100 microns, or 100-1000 microns). Such chiplets can be made in a native source semiconductor wafer (e.g., a silicon wafer) having a process side and a back side used to handle and transport the wafer using lithographic processes. Micro-devices **20** can be formed using lithographic processes in an active layer on or in the process side of source substrate **10**. Methods of forming such structures are described, for example, in U.S. Pat. No. 8,889,485. According to some embodiments of the present disclosure, source substrates **10** can be provided

with micro-devices **20**, sacrificial layer **12** (a release layer), sacrificial portions **14**, and tethers **40** already formed, or they can be constructed as part of a process in accordance with certain embodiments of the present disclosure.

(75) In certain embodiments, micro-devices **20** can be constructed using foundry fabrication processes used in the art. Layers of materials can be used, including materials such as metals, oxides, nitrides and other materials used in the integrated-circuit art. Micro-devices **20** can have different sizes, for example, less than 1000 square microns or less than 10,000 square microns, less than 100,000 square microns, or less than 1 square mm, or larger. Micro-devices **20** can have, for example, at least one of a length, a width, and a thickness of no more than 500 microns (e.g., no more than 250 microns, no more than 100 microns, no more than 50 microns, no more than 25 microns, or no more than 10 microns). Micro-devices **20** can have variable aspect ratios, for example at least 1:1, at least 2:1, at least 5:1, or at least 10:1. Micro-devices **20** can be rectangular or can have other shapes.

(76) Tethers **40** can comprise any suitable tether material and can incorporate one or more layers, for example one or more layers similar to or the same as those layer(s) of micro-device **20**, for example comprising electrode material, dielectric(s), or encapsulation layer(s), including resins, silicon oxides, silicon nitrides, or semiconductors. Tethers **40** can be constructed be depositing (e.g., by evaporation or sputtering) material such as oxide, nitride, metal, polymer, or semiconductor material, and patterning the material, for example using photolithographic methods and materials, such as pattern-wise exposed and etched photoresist.

(77) Source substrate **10** can be any useful substrate with a surface suitable for forming or having patterned sacrificial layers **12**, sacrificial portions **14**, anchor portions **16**, and forming or disposing micro-devices **20**. Source substrate **10** can comprise glass, ceramic, polymer, metal, quartz, or semiconductors, for example as found in the integrated circuit or display industries. Sacrificial portion **14** can be a designated portion of sacrificial layer **12**, for example an anisotropically etchable portion, for example designated by virtue of etchant applied to source substrate **10** is exposed to it relative to other portions of source substrate **10**, or a differentially etchable material from sacrificial layer **12**, for example a buried oxide or nitride layer, such as silicon dioxide. A surface of source substrate **10** surface can be substantially planar and suitable for photolithographic processing, for example as found in the integrated circuit or MEMs art. Source substrate **10** can be chosen, for example, based on desirable growth characteristics (e.g., lattice constant, crystal structure, or crystallographic orientation) for growing one or more materials thereon. In some embodiments of the present disclosure, source substrate **10** is anisotropically etchable. For example, source substrate **10** can be a monocrystalline silicon substrate with a {100} orientation. An anisotropically etchable material etches at different rates in different crystallographic directions, due to reactivities of different crystallographic planes to a given etchant. For example, potassium hydroxide (KOH) displays an etch rate selectivity 400 times higher in silicon [100] crystal directions than in silicon [111] directions. In particular, silicon {100} is a readily available, relatively lower cost monocrystalline silicon material. Moreover, in some embodiments, micro-devices **20** made on or in a silicon {100} crystal structure can have less stress and therefore less device bowing after release.

(78) The present disclosure provides, inter alia, a structure and method for improving the release of a micro-transfer-printable micro-device structure **90** from source substrate **10** (source wafer) by etching, in particular where sacrificial layer **12** in source substrate **10** comprises an anisotropically etchable crystalline semiconductor material such as silicon {100}. Patterned sacrificial layer **12** defines sacrificial portions **14** comprising a sacrificial material laterally spaced apart by anchor portions **16**. Anchor portions **16** can be a non-sacrificial portion of sacrificial layer **12**, can be disposed over sacrificial layer **12**, or both. Anchor portions **16** can comprise a non-sacrificial portion of sacrificial layer **12** and, optionally, material deposited on sacrificial layer **12**, for example the same material deposited to form micro-devices **20** in a common step or an

encapsulating material in a common deposition step. A micro-device **20** is disposed entirely and completely over each sacrificial portion **14** and is physically connected to anchor portion **16** by tether **40**. An encapsulation layer comprising an encapsulation material can encapsulate any one or combination of micro-devices **20**, tether **40**, and anchor portion **16**. In some embodiments, an encapsulation layer forms tether **40** or a portion of tether **40** so that tether **40** comprises the encapsulation layer or the encapsulation material. In some embodiments, the encapsulation layer forms anchor portion **16** or a portion of anchor portion **16**.

(79) According to some embodiments of the present disclosure, etch hole **30** is disposed in micro-device **20** and extends through micro-device **20**. Etch hole **30** can be a hole, for example a shaped hole. Etch holes **30** can be formed by pattern-wise etching micro-device **20**, for example using deposited (e.g., by spin or curtain coating) patterned photoresists such as are found in the integrated circuit arts. Once etch hole **30** is present in micro-device **20**, an etchant, for example a liquid etchant, can be disposed over micro-device **20** and tether **40** on a side of over micro-device **20** and tether **40** opposite sacrificial portion **14** and pass through etch hole **30** to contact and etch sacrificial portion **14**. Etch hole **30** can have various shapes and is necessarily smaller than micro-device **20**. In some embodiments, etch hole **30** has a shape approximately similar or geometrically similar (e.g., having the same relative proportions) as micro-device **20**.

(80) In some embodiments, etchants can also be applied to sacrificial portion **14** around a perimeter of micro-devices **20** and tethers **40** to release micro-devices **20** from source substrate **10** (in addition to through etch holes **30**). In some embodiments of a micro-transfer printing materials system, anisotropically etchable sacrificial portions **14** take a longer time to etch and release micro-devices **20** than is desired, leading to etching damage to micro-devices **20** and tethers **40** and reducing manufacturing throughput. These issues are addressed, according to embodiments of the present disclosure, by providing etch hole **30** in micro-device **20**. Etch hole **30** provides access to etchants and can have convex angles that enable fast and efficient etching beneath micro-device **20** and, optionally tether **40**. Sacrificial portion **14** etchants pass through etch hole **30** to attack sacrificial portion **14** beneath micro-device **20** at convex corners, decreasing the time necessary to fully release micro-device **20** and tether **40** from source substrate **10** and prepare micro-device **20** for micro-transfer printing with a stamp.

(81) As is understood by those skilled in the art, the terms “over” and “under” are relative terms and can be interchanged in reference to different orientations of the layers, elements, and substrates included in the present disclosure. For example, a first layer on a second layer, in some implementations means a first layer directly on and in contact with a second layer. In other implementations, a first layer on a second layer includes a first layer and a second layer with another layer therebetween.

(82) Having described certain implementations of embodiments, it will now become apparent to one of skill in the art that other implementations incorporating the concepts of the disclosure may be used. Therefore, the disclosure should not be limited to certain implementations, but rather should be limited only by the spirit and scope of the following claims.

(83) Throughout the description, where apparatus and systems are described as having, including, or comprising specific components, or where processes and methods are described as having, including, or comprising specific steps, it is contemplated that, additionally, there are apparatus, and systems of the disclosed technology that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the disclosed technology that consist essentially of, or consist of, the recited processing steps.

(84) It should be understood that the order of steps or order for performing certain action is immaterial so long as operability is maintained. Moreover, two or more steps or actions in some circumstances can be conducted simultaneously. The disclosure has been described in detail with particular reference to certain embodiments thereof, but it will be understood that variations and modifications can be effected within the spirit and scope of the claimed invention.

PARTS LIST

(85) A cross section line B cross section line D direction/distance W.sub.T tether width W.sub.M micro-device width W.sub.E etch-hole width L.sub.M micro-device length **10** source substrate/source wafer **11** source substrate surface **12** sacrificial layer **14** sacrificial portion **15** gap/cavity **16** anchor portion/anchor **18** source substrate flat **20** micro-device **22** micro-device edge/edge side **24** top side **26** bottom side **30** etch hole **32** etch-hole edge **33** etch-hole protective coating **34** pinned etch front/pinned etch plane **36** concave corner **37** straight line **38** orthogonal line segment angle **40** tether **42** structure tether **50** dielectric structures **52** intermediate substrate **60** target substrate **70** antenna **72** micro-heater **74** micro-fluidic structure/nano-fluidic structure **76** power device **78** channel **90** micro-device structure **92** micro-device structure **94** micro-device structure/micro-device system **95** intermediate structure **100** provide source substrate step **110** provide micro-device step **120** provide etchant step **130** etch source substrate step **140** provide stamp step **150** contact stamp to micro-device step **160** remove stamp and micro-device step **170** provide target substrate step **180** press micro-device to target substrate with stamp step **190** remove stamp step

Claims

1. A micro-device structure, comprising: a micro-device having a top side and a bottom side and comprising one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends entirely through an interior portion of the micro-device substrate from the top side to the bottom side; one or more anchors; and wherein any straight line parallel to an edge side of the micro-device drawn from a concave corner on a first edge side of the micro-device to a second edge side of the micro-device opposite the first edge side contacts at least one of the one or more etch holes.
2. The micro-device structure of claim 1, wherein the concave corner is defined by the micro-device.
3. The micro-device structure of claim 1, comprising one or more tethers extending from the micro-device, wherein the concave corner is defined by the micro-device and/or the one or more tethers.
4. A micro-device structure, comprising: a micro-device having a top side and a bottom side and comprising two or more concave corners and one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends entirely through an interior portion of the micro-device substrate from the top side to the bottom side; one or more anchors; and one or more tethers physically connecting the micro-device to the one or more anchors, wherein any orthogonal line segment angle drawn from a first concave corner defined by the micro-device and/or the one or more tethers and disposed on a first edge side of the micro-device to a second concave corner defined by the micro-device and/or the one or more tethers and disposed on a second edge side of the micro-device adjacent to the first edge side contacts at least one of the one or more etch holes.
5. A micro-device having a top side and a bottom side and comprising one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends entirely through an interior portion of the micro-device substrate from the top side to the bottom side, wherein the one or more etch holes comprise a first etch hole extending into the micro-device from a first edge side, a second etch hole extending into the micro-device from a second edge side that is opposite the first edge side, and a third etch hole extending into the micro-device from a third edge side different from the first edge side and from the second edge side, wherein the third etch hole comprises a first portion that intersects the third edge side and a second portion that intersects an end of the first portion at a point other than an end of the second portion.
6. The micro-device of claim 5, wherein the first portion bisects the second portion.
7. The micro-device of claim 5, wherein the first etch hole and the second etch hole are parallel.
8. The micro-device of claim 5, wherein the first etch hole and the second etch hole are collinear.

9. The micro-device of claim 5, wherein the first portion extends into the micro-device past the first etch hole and the second etch hole.
10. A micro-device having a top side and a bottom side and comprising one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends entirely through an interior portion of the micro-device substrate from the top side to the bottom side, and wherein the one or more etch holes comprise a plurality of etch holes disposed in a symmetric arrangement, the symmetric arrangement having a multifold symmetry and comprising a unit cell comprising a first portion of a first etch hole extending outward from a center of the micro-device, a second etch hole that is an interior etch hole and disposed parallel to an edge side of the micro-device, and a third etch hole that is an interior etch hole and that is co-linear with a center of the micro-device.
11. The micro-device of claim 10, wherein the symmetric arrangement has a two-fold symmetry or a four-fold symmetry.
12. The micro-device of claim 10, wherein the first etch hole has a cross shape.
13. The micro-device of claim 10, wherein the second etch hole and the third etch hole are straight etch holes.
14. The micro-device of claim 10, wherein the third etch hole is collinear with both the center of the micro-device and a corner of the micro-device.
15. A micro-device having a top side and a bottom side and comprising one or more etch holes, wherein each of the one or more etch holes is an etch hole that extends entirely through an interior portion of the micro-device substrate from the top side to the bottom side, and wherein the one or more etch holes comprise a first etch hole comprising a first portion and a second portion, wherein the first portion and the second portion intersect at respective interior points and a center of the first etch hole is disposed closer to a center of the micro-device than any edge side of the micro-device.
16. The micro-device of claim 15, wherein the first portion and the second portion each intersect the center of the micro-device.
17. The micro-device of claim 15, wherein the first portion and the second portion form a cross.
18. The micro-device of claim 15, wherein the first portion has a length equal to a length of the second portion.
19. The micro-device of claim 15, wherein the first etch hole is the only etch hole.